

Software Requirements Specification

Project: Domain & SSL Certificate Expiry Alert System - BPS #47
Student: Sujay Hegde | SRN: PES1UG24CS478 | Institution: PES University
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1. Introduction

1.1 Purpose

Specify the behavior, interfaces, constraints, and acceptance procedures for a service that checks domain registration and SSL/TLS expiry and warns responsible administrators. The intended readers are SysAdmins, Security Officers, the developer, and the lab evaluator.

1.2 Scope

The service maintains a monitored asset list, performs daily TLS and WHOIS checks, issues threshold-based alerts, tracks acknowledgment and escalation, and emails monthly summaries. It assists renewal decisions; it does not buy domains, renew certificates, modify DNS records, or replace a certificate authority.

1.3 Terms and references

An asset is a domain registration or a configured TLS endpoint. An expiry alert is a notification tied to an asset, expiry date, and threshold. Acknowledgment records responsibility for an alert; it does not mean renewal occurred. WHOIS provides registration information; TLS is used to obtain endpoint certificate information. FR, NFR, RTM, and WBS mean functional requirement, non-functional requirement, requirements traceability matrix, and work breakdown structure.

The baseline is BPS #47 and Requirements.md. Use cases, the acknowledgment flow, and the RTM provide supporting models. Folder 2's coffee kiosk is an independent architecture exercise from the Lab 3 handout.

2. Overall description

2.1 Product perspective

Proposed BPS #47 design: a browser dashboard calls an authenticated application service. A scheduler starts daily checks and escalation evaluations. TLS and WHOIS adapters supply expiry information to the alert engine. A persistent repository stores assets, scan results, alerts, acknowledgment events, and delivery records. A notification adapter sends alerts and reports. These modules are a proposed project design; they are not the coffee-kiosk components.

2.2 Actors and privileges

Actor	Responsibilities	Permissions
SysAdmin	Maintain monitoring configuration; inspect and acknowledge alerts; read reports	Add/edit/remove authorized assets; acknowledge authorized alerts
Security Officer	Review escalations and monthly summaries	Read assigned asset/alert information; no monitoring-list mutation
External Scheduler	Trigger time-based background work	Start daily scans, check escalation deadlines, request monthly reports through a service identity

2.3 Operating assumptions and constraints

The eventual service needs outbound DNS, WHOIS, TLS, and mail access, an HTTPS dashboard, durable storage, and an accurate clock. Lookup responses can be missing, malformed, rate-limited, or unavailable. No assumption is made that every registry returns WHOIS expiry data. Unknown expiry is displayed as unknown, with a diagnostic, rather than as healthy. Network timeouts, retry budgets, and concurrency limits must be configured before benchmarking.

Times are stored in UTC. Certificate thresholds use remaining UTC time; a date-only registration expiry must retain its source precision. The prototype uses injected UTC fixtures; production WHOIS parsing and scheduling remain future integration work.

3. External interfaces

3.1 Dashboard

Provide a monitored-assets list, add/edit/remove form, latest scan status, expiry warnings, an acknowledgment action, escalation status, and report access. Show unavailable results and the timestamp of the last successful check. Require authentication and verify permission on the server for each operation.

3.2 Lookup interfaces

TLS adapter input: hostname and endpoint port, with timeout and server-name indication. Output: certificate expiry and validation status, or an explicit error. An expired or invalid certificate must not silently become a successful result. WHOIS adapter input: domain. Output: normalized registration expiry and source metadata, or an explicit unavailable/parse error. Keep raw response parsing separate from alert policy.

3.3 Notification and storage interfaces

Notifications carry asset identity, expiry date, threshold, alert ID, and relevant contacts. Delivery uses a configured mail provider; credentials are excluded from reports and source control. Failed delivery is recorded and retried within a bounded budget. Storage mutations use transactions so acknowledgment and escalation cannot both win an uncoordinated race. Use a stable alert key based on asset, expiry, and threshold to prevent duplicate daily notifications.

4. Functional requirements

The following summarizes the existing five requirements without introducing additional FR identifiers. The complete descriptions, priorities, acceptance criteria, and rationales remain in the baseline requirements table.

ID	Required behavior	Priority	Acceptance focus
FR-001	Daily TLS checks; certificate alerts at 30, 15, and 3 days before expiry; expired certificates reported	High	15-day fixture produces an alert; expiry is never silently ignored
FR-002	Daily WHOIS checks; registration alerts at 45, 30, and 7 days	High	Correct threshold warnings; lookup failure recorded without crashing
FR-003	Authorized acknowledgment; Security Officer escalation for high-priority assets unacknowledged after 48 hours	Medium	Before-deadline acknowledgment suppresses escalation; overdue alert escalates once

ID	Required behavior	Priority	Acceptance focus
FR-004	Secure dashboard supports adding, modifying, and removing domains and endpoints	High	Current registry drives the next scan; removed assets are excluded
FR-005	Generate monthly all-asset summary and email stakeholders	Low	First-of-month report includes every monitored asset and unknown/error states

5. Non-functional requirements

ID	Target	Measurement
NFR-001	Scan 1,000 configured domains/endpoints in less than 180 seconds; HTTPS and role-based dashboard access	Record machine, concurrency, network profile, timeout budget, asset mix, and elapsed wall time. Include timeout records in the scan result. Exercise unauthenticated and unauthorized requests against deployed HTTPS routes.
NFR-002	At least 99.9% monthly service availability; at most three exponential-backoff retries after a failed initial WHOIS/DNS attempt	Measure health over a full calendar month. Inject transient/persistent failures and verify increasing retry delays, bounded attempts, error logs, and continued processing of other assets.

These are acceptance targets, not measured achievements. A simulated unit test does not establish performance, HTTPS deployment, or monthly uptime.

6. Data model and business rules

Entity	Principal fields	Integrity rule
Monitored asset	ID, domain, endpoint port, owner, security contact, priority, enabled flag	Unique configured identity; authorized mutation only
Scan result	Asset ID, kind (TLS/WHOIS), checked time, expiry, status, error	Preserve latest success and most recent failure separately
Expiry alert	ID, asset, kind, expiry, threshold, created time, state	Stable unique key prevents repeated threshold alerts
Acknowledgment	Alert ID, actor, timestamp	Authorized actor; atomic state transition
Escalation/delivery	Alert/report ID, recipient, state, attempts, timestamp	Record failures; do not claim delivery before provider success
Monthly report	Month, generated time, asset snapshots, delivery state	One logical report per reporting month; retries reuse report identity

An expired asset raises an expiry warning regardless of the normal pre-expiry threshold. A lookup failure is separate from an expiry alert. Asset removal prevents future scans; audit records may be retained according to an agreed retention policy. Renewed expiry dates create a new alert cycle. Acknowledged alerts stop pending escalation but do not suppress future renewal-cycle alerts.

7. Workflows and failure handling

Daily check: scheduler selects current enabled assets; adapters perform bounded lookups; scan results are recorded; alert policy checks the appropriate threshold; notification dispatch is recorded. One lookup failure must not abort unrelated assets.

Acknowledgment: authenticated SysAdmin opens the alert, permission is checked, state changes atomically, and an audit event is written. At or after 48 hours, the scheduler selects still-unacknowledged high-priority alerts and attempts escalation. Production storage must prevent duplicate escalation under concurrent workers.

Monthly report: take a consistent current-asset snapshot, include success/unknown/error states, generate the summary, and dispatch to configured stakeholders. Persist a monthly identity so a restarted job cannot create duplicate reports.

8. Verification and release criteria

Run the 14 verification procedures in the RTM. Unit fixtures may validate thresholds, bounded retries, acknowledgment, registry changes, and report completeness. Separate integration tests must validate actual TLS/WHOIS/mail adapters, database transactions, daily/monthly scheduling, and dashboard authorization. The performance benchmark and monthly health measurement require their own recorded evidence.

Release requires all five FRs to pass their acceptance procedures, both NFRs to meet their measured targets, and all unresolved integration issues to be recorded. Captured Jira defects BB-1 through BB-4 remain To Do unless the actual Jira records are updated with verified results.

9. Open decisions and current implementation status

Production database, mail provider, WHOIS parser, deployment environment, benchmark profile, retention period, and scheduler timezone are to be selected during implementation. The folder-5 starter is a local core prototype using injected lookup and notification adapters and in-memory state. It does not provide the production dashboard, live scanning, persistent transactions, scheduling, or operational monitoring. Copilot provenance requires an actual Copilot session or existing Copilot-generated repository.

Work Breakdown Structure and Execution Steps

Sujay Hegde | PES1UG24CS478 | BPS #47 - Domain & SSL Certificate Expiry Alert System

The hierarchy and work-package dictionary describe the project's work. Status reflects available artifacts, not an invented schedule or completed production implementation. Sujay Hegde is the owner of each work package.

1. Hierarchical WBS

- 1.0 Domain & SSL Certificate Expiry Alert System
 - 1.1 Requirements and planning
 - 1.1.1 Define actors, scope, five FRs, and two NFRs
 - 1.1.2 Model use cases and acknowledgment flow
 - 1.1.3 Produce RTM and SRS
 - 1.1.4 Organize Jira epics, backlog, and defect evidence
 - 1.2 Secure configuration and persistence
 - 1.2.1 Authentication and role authorization
 - 1.2.2 Asset registry and dashboard CRUD
 - 1.2.3 Durable records, unique alert keys, and audit events
 - 1.3 Monitoring and notification
 - 1.3.1 Daily scheduler and TLS adapter
 - 1.3.2 WHOIS adapter, parsing, timeouts, and retries
 - 1.3.3 Threshold alert lifecycle and acknowledgment
 - 1.3.4 48-hour escalation and notification delivery
 - 1.3.5 Monthly report generation and email
 - 1.4 Verification and operations
 - 1.4.1 Functional tests, 1,000-asset benchmark, HTTPS security checks
 - 1.4.2 Fault isolation, recovery, and monthly availability monitoring
 - 1.5 Assignment documentation and evidence
 - 1.5.1 Separate coffee-kiosk architectural exercise
 - 1.5.2 GitHub and Jira evidence indexing
 - 1.5.3 Actual Copilot generation and evidence capture
 - 1.5.4 Final README, PDFs, and submission review

2. Work-package dictionary

Package	Output / completion criterion	Requirement	Depends on	Current status
WP-1.1	Approved requirements, actors, use cases, flow, RTM, SRS	All seven	None	Documents prepared
WP-1.2	Jira setup and defect evidence indexed with source pages	All seven	WP-1.1	Captured setup evidence available; sprint completion not asserted
WP-2.1	Login/session security and authorization checked on every protected route	FR-003, FR-004, NFR-001	WP-1.1	Starter role guard only; production auth planned

Package	Output / completion criterion	Requirement	Depends on	Current status
WP-2.2	Asset CRUD dashboard backed by durable registry; removed asset excluded from scans	FR-004	WP-2.1	Starter registry fixture only; dashboard/persistence planned
WP-2.3	Transactional alert/scan/audit storage; duplicate and race tests pass	FR-001 to FR-005	WP-2.2	Planned
WP-3.1	Actual TLS adapter and daily job detect expiry and validation failure	FR-001	WP-2.2, WP-2.3	Injected fixtures available; live adapter/scheduler planned
WP-3.2	WHOIS parser and bounded retries handle unavailable registry data	FR-002, NFR-002	WP-2.2	Injected fixtures available; live adapter planned
WP-3.3	Alert thresholds and acknowledgment state transitions tested	FR-001, FR-002, FR-003	WP-3.1, WP-3.2	Starter logic and fixtures available; durable integration planned
WP-3.4	Escalation at 48h; actual mail delivery and retry evidence	FR-003	WP-3.3	Starter deadline logic available; mail integration planned
WP-3.5	All-asset summary; scheduled monthly delivery, identity, and retry checks	FR-005	WP-3.1, WP-3.2, WP-3.4	Starter report fixture only; scheduling/mail planned
WP-4.1	TC-001 to TC-012 executed; benchmark below 180s and HTTPS verified	Five FRs, NFR-001	WP-2.1 to WP-3.5	Core unit tests available; integration/security/performance planned
WP-4.2	Failure isolation, recovery, and measured 99.9% monthly availability	NFR-002	WP-3.2, WP-4.1	Starter failure fixture available; operational measurement planned
WP-5.1	Coffee-kiosk diagram and one-page justification exported	Separate Lab 3 handout	None	Available in folder 2
WP-5.2	Real repository/history and Jira screenshots indexed	Assignment evidence	WP-1.2	Available in folder 3

Package	Output / completion criterion	Requirement	Depends on	Current status
WP-5.3	Run GitHub Copilot on project code; retain prompt/output and real screenshot or repository link	Assignment item 5	WP-1.1	Pending actual Copilot session; Codex starter supplied
WP-5.4	All deliverables linked; PDFs readable; no missing or falsely attributed evidence	Assignment checklist	WP-5.1 to WP-5.3	Documentation prepared; Copilot evidence outstanding

3. Execution steps and review gates

1. Review the seven baseline requirements and actor permissions. Resolve production decisions in SRS section 9 before selecting adapters.
2. Implement authentication, registry CRUD, and durable storage. Gate: unauthorized mutation rejected and removed domains excluded from the next scan.
3. Integrate TLS and WHOIS adapters and the daily scheduler. Gate: controlled real endpoints produce correct expiry/error records with bounded attempts.
4. Integrate alert notifications, acknowledgment, and 48-hour escalation. Gate: thresholds are correct; duplicate daily notifications and acknowledgment/escalation races are prevented.
5. Integrate monthly report scheduling and mail. Gate: every current asset is represented, including failed checks, and report retries do not duplicate logical reports.
6. Execute the RTM verification plan. Gate: functional integration evidence, the defined performance benchmark, and HTTPS/role checks are recorded separately from prototype unit tests.
7. Deploy health monitoring and observe availability over a complete month. Gate: measured availability meets NFR-002; failures and recovery procedures are documented.
8. Complete the actual Copilot session, capture evidence, verify all links and exported PDFs, and review the assignment checklist before submission.

4. Risk and mitigation

Risk	Effect	Planned mitigation / verification
WHOIS expiry absent or rate-limited	False healthy status or missed registration warning	Preserve unknown/error state; bounded retries; TC-004, TC-013
Invalid or expired TLS certificate	Handshake failure hides expiry	Explicit validation failure; controlled endpoint integration in TC-002
Duplicate job execution	Repeated warnings or escalations	Stable keys and durable atomic state; repeat/concurrent-worker tests
Acknowledgment races escalation	Unnecessary security notification	Transactional conditional update; acknowledgment and deadline integration tests
Provider latency exceeds benchmark	NFR-001 fails	Record latency profile; bounded concurrency and timeout budget; TC-011

Risk	Effect	Planned mitigation / verification
Screenshot or generated-code attribution unsupported	Incomplete evidence submission	Preserve provenance; capture actual pages; retain actual Copilot output before marking WP-5.3 complete

No calendar deadlines or production completion percentages are asserted. The existing Scrum report's 26 planned points describe that captured sprint; they are not an estimate for this complete WBS.